

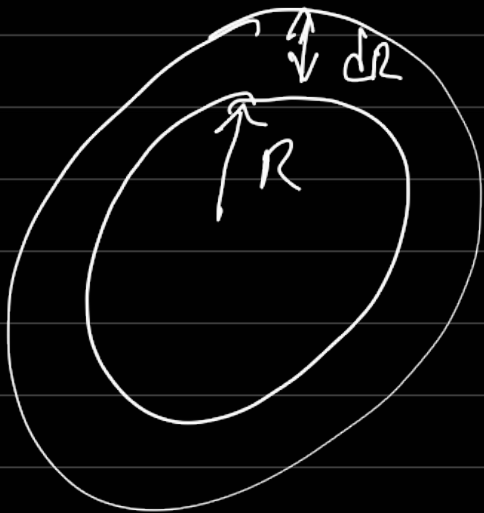
$$\textcircled{1} I(R) = I_0 e^{-R/R_h}$$

$$\begin{aligned} \mu(R) &= -2.5 \log(I) \\ &= -2.5 \frac{\ln(I)}{\ln(I_0)} = -\frac{2.5}{\ln(I_0)} \left(\ln(I_0) - \frac{R}{R_h} \right) \end{aligned}$$

$$\mu(R) = -\frac{2.5}{\ln(I_0)} \ln(I_0) + \frac{2.5}{\ln(I_0)} \frac{R}{R_h}$$

$$\frac{2.5}{\ln(I_0)} = 1.086$$

$$\mu(R) = \mu_0 + 1.086 \frac{R}{R_h}$$



$$I \sim \frac{L}{A}$$

$$dL = 2\pi R dR I$$

$$L(<R) = \int_0^R 2\pi R I_0 e^{-R/R_h} dR$$

$$L(<R) = 2\pi I_0 R_h^2 \left(1 - e^{-R/R_h} \left(1 + \frac{R}{R_h} \right) \right)$$

R_h : encloses half light

$$L(R_h) = \pi I_0 R_h^2 = 2\pi I_0 R_h^2 \left(1 - e^{-1} \left(1 + \frac{R_h}{R_h} \right) \right)$$

$$\frac{1}{2} = \left(1 - e^{-R} (1 + R) \right)$$

$$1 - e^{-R} (1 + R)$$

$$\frac{1}{2} = \sum \frac{R^n}{n!} (-1)^n (1 + R)$$

...

$$\textcircled{2} \mu_0 = 21.65, M_B = 13.7$$

$$\mu(R) = \mu_0 + 1.086 \frac{R}{R_u}$$

$$\mu(R_{25})$$

